21 $5c + d = c + 4d$

(a) Find the ratio $c : d$

$5c + d = c + 4d$ gives $4c = 3d$.

Hence $c/d = 3/4$, so $c : d = 3 : 4$.

2 marks total
Collect c and d
3 : 4

(2)

$6x^2 = 7xy + 20y^2$ where $x > 0$ and $y > 0$

(b) Find the ratio $x : y$

$6x^2 - 7xy - 20y^2 = 0$.

Factor: $(2x - 5y)(3x + 4y) = 0$.

Since $x, y > 0$, $3x + 4y > 0$.

Thus $2x = 5y$ and $x : y = 5 : 2$.

3 marks total
Rearrange quadratic
Factor correctly
Positive ratio 5 : 2

(3)

(Total for Question 21 is 5 marks)

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